

Project 1.1.7 Tug-of-War

PLTW

TUG-OF-WAR



Tow Strength



Precision and Accuracy



Initial Competition



Conclusion

Tow Strength

GOALS	MATERIALS	RESOURCES
• Design and construct a prototype to solve an engineering design challenge. • Conduct quantitative tests to inform iteration throughout an engineering design challenge. • Differentiate between accuracy and precision in measurement. • Apply knowledge of mass, friction, and gearing to an engineering design problem.		
GOALS	MATERIALS	RESOURCES
• VEX® V5 POE Kit • Masking Tape • Vernier® Go Direct® Force and Acceleration Sensor		

GOALS	MATERIALS	RESOURCES
• <u>Tug-of-War Code</u> • <u>Project 11.7 Tug-of-War Rubric</u> • <u>VEX® V5 Build Support Guides</u> • <u>Collaboration</u> • <u>Conflict Resolution</u> • <u>Vernier Graphical Analysis®</u> • <u>Vernier® Sensors and Probes</u> • <u>Project 11.7 Tug-of-War (Downloadable PDF)</u>		

Many farms require equipment that is capable of pulling heavy loads. Whether it is hay bales, harvesting machines, or trailers full of horses, the ability to pull heavy loads is essential for some farmers. A horse trailer can weigh close to 8,000 pounds!



In this project, you'll apply your engineering knowledge using your VEX® V5 POE Kit to design a machine that can pull with as much force as possible.

Your Challenge: Tug-of-War

You and a partner will design a vehicle that can pull with the greatest force possible. Your vehicle can include only two VEX® V5 Smart Motors. You will use the Vernier® Go Direct® Force and Acceleration sensor to measure how much force your robot pulls with. You will also measure your robot's success by competing in tug-of-war battles with classmates.

1

Follow your teacher's instructions to find a partner.

2

Construct an initial prototype. For help constructing your tug-of-war machine, use the [**VEX® V5 Build Support Guides**](#).

Collaboration and Conflict Resolution: You and your partner may have different ideas about how to

construct your tug-of-war machine. Use the [Collaboration](#) and [Conflict Resolution](#) resources to support you in successfully working together to complete the tug-of-war challenge.

3

Download the [Tug-of-War code](#) onto your VEX V5 Brain. This code makes your robot drive forward until the code ends.

NOTE: This code is designed to run two motors plugged into Ports 1 and 2. Make sure your motors are plugged into these ports; otherwise, the code will not work. If you want to change the direction your robot drives, switch the port each motor is plugged into and the drive direction will switch.

- Use the Vernier Go Direct Force and Acceleration Sensor to measure how many Newtons of force your tug-of-war machine pulls with.

Vernier: If you need further guidance or support, review the [Vernier Graphical Analysis®](#) and [Vernier® Sensors and Probes](#) resources.

- Hit the sensor to the back of your robot. To get the best readings, select a square hole toward the center of a C-Channel or similar metal assembly piece as shown in Figure 1. Once attached, hold the sensor firmly in place.

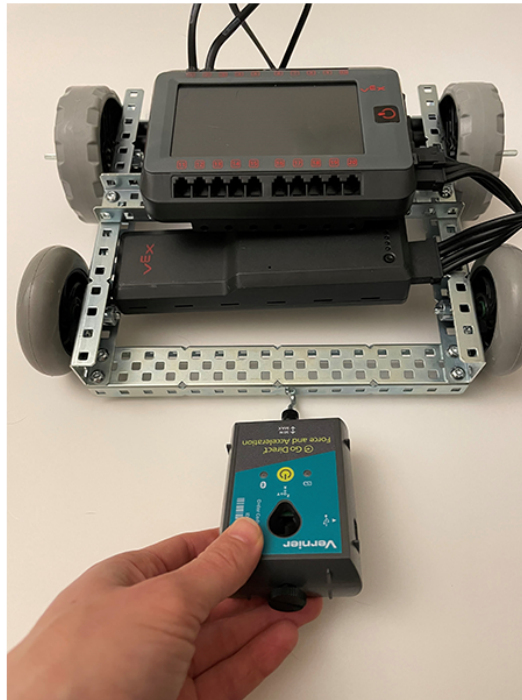


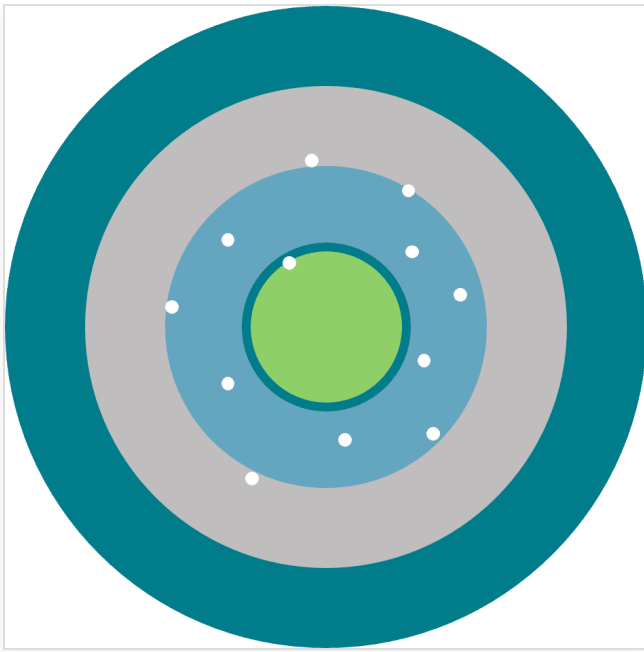
Figure 1. Example Force Measurement Setup

- Have one partner start collecting data in the Graphical Analysis application while the other partner runs the Tug-of-War code on the VEX V5 Robot Brain.
 - View the graph of your data and determine the force exerted by your tug-of-war machine.
- Perform the sequence for step 3 several times to eliminate any errors in measuring.

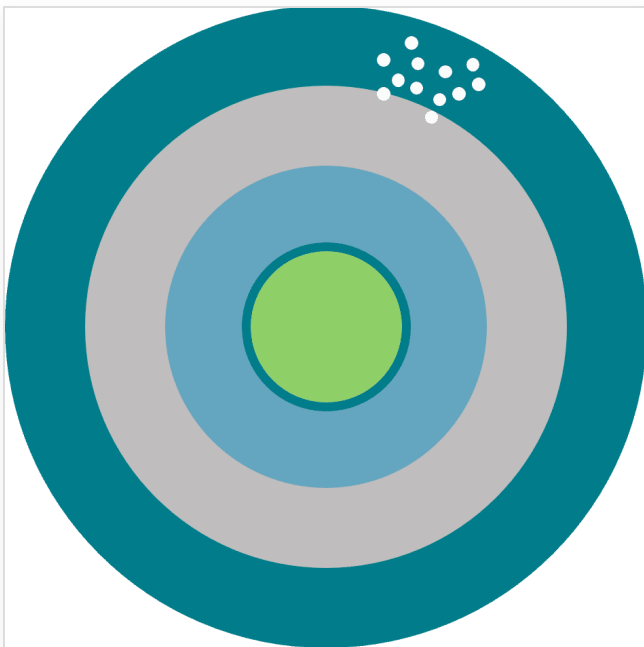
Precision and Accuracy

To evaluate the performance of your team's tug-of-war machine, consider the *accuracy* and *precision* of the measurement data you collected using the Vernier® Go Direct® Force and Acceleration sensor.

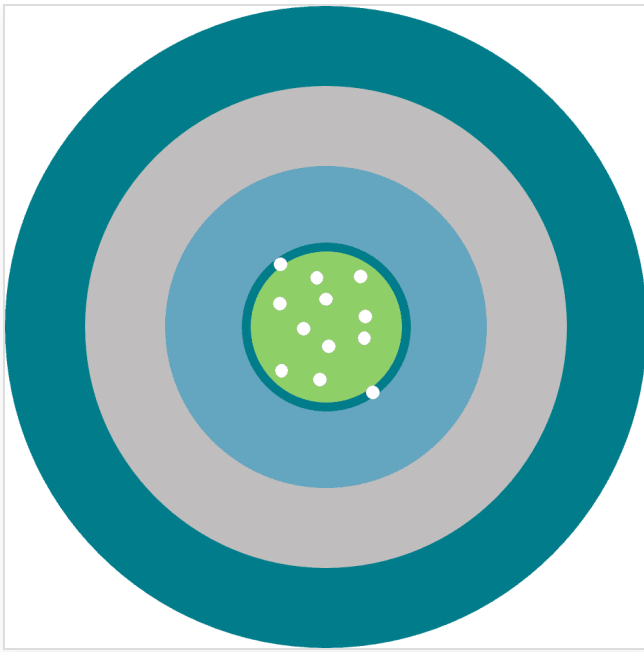
The *accuracy* of a measurement is dependent on the instrument's calibration to a set standard. Poor accuracy often stems from procedural or equipment flaws. Conversely, the *precision* of a measurement is dependent on the capabilities of a measurement device as well as how it is used. Toggle through the target images to view examples of precision and accuracy.



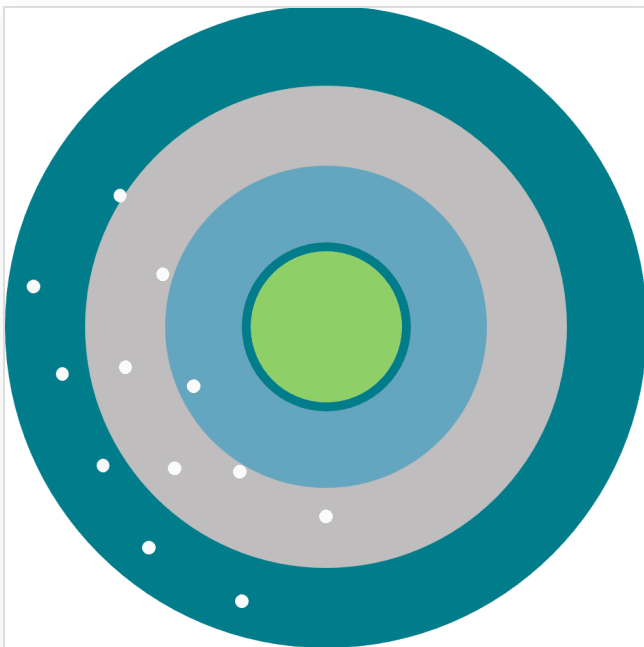
accurate, but not precise



precise, but not accurate



both accurate and precise



neither accurate or precise

Measurements always involve uncertainty due to random and systematic errors. As such, a measurement is the best *estimate* of a quantity. When taking a measurement, be sure to determine how accurate and precise the value may be.

Turn and Talk



Is poor precision usually associated with random or systematic error?

Initial Competition

4

Engage in an initial competition.

- Connect your robot to another group's robot using a 1x25 Steel Bar (Figure 2) so that your robot pulls in the opposite direction of the other robot when you run the code.
- Position the midpoint of the steel bar perpendicular to the midpoint of the tug-of-war centerline (shown in Figure 3).
- When both teams are ready, simultaneously run the Tug-of-War code on each robot and see what happens. Whichever robot pulls the other one across the centerline, wins!

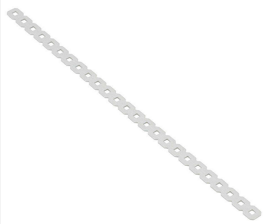


Figure 2. 1x25 Steel Bar for Tug-of-War

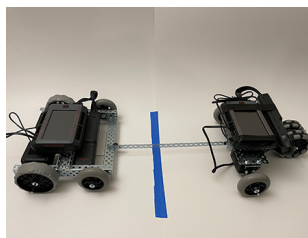


Figure 3. Example Tug-of-War Competition Setup



Engineering Notebook

Be sure to record your results. Did your vehicle pull the competing vehicle over the line? Was your vehicle pulled over the line? Was it a tie?

5

Reflect on your design. Why do you think it performed the way it did? What factors affected your vehicle's ability to maximize its pulling or pull force? In particular, consider the following questions:

- How could you use gears to maximize the pull force of your robot?
- How does your robot's mass impact how much weight it can pull?
- How might friction impact your robot's tugging ability?

6

Iterate on your initial prototype and make sure to incorporate the concepts of mass, gearing, and friction.

VEX® V5 Build Support Guides: For help incorporating gears into your design, reference the [VEX® V5 Build Support Guides](#).

7

For each iteration, use the Vernier® Go Direct® Force and Acceleration sensor to measure the pull force exerted by your tug-of-war machine, and complete at least one tug-of-war battle with another group. Record your results and note how your iterative changes impacted your robot's pulling performance.

8

Use the [Project 11.7 Tug-of-War Rubric](#) to evaluate your project performance.

Conclusion

Question 1

How did you apply your understanding of energy, work, and power to your design?

Question 2

Did your robot reach maximum motor power? How do you know?

Question 3

If you repeated this challenge, what would you do differently? Why?

Question 4

Outside of farming and tug-of-war, list examples of other situations where devices or machines are optimized to pull with a lot of force.